Ref. Ares(2020)3139869 - 17/06/2020

ESSnet Big Data II

Grant Agreement Number: 847375-2018-NL-BIGDATA

https://webgate.ec.europa.eu/fpfis/mwikis/essnetbigdata

https://ec.europa.eu/eurostat/cros/content/essnetbigdata_en

Workpackage J

Innovative Tourism Statistics

Deliverable J4: Technical Report.

Final version, 16.06.2020

Prepared by:

WPJ Team

Workpackage Leader:

Marek Cierpiał-Wolan (Statistics Poland, Poland)
M.Cierpial-Wolan@stat.gov.pl
telephone : + 48 17 85 35 210 ext. 311
mobile phone : + 48 515 037 640

Table of contents

Executive summary 3
Glossary 4
1. Introduction 5
1.1 Technical review 5
1.2 Design requirements 8
2. Prototype description 9
2.1 Overview 9
2.2 Prototype components 11
2.2.1 Data gathering 12
2.2.2 Pre-processing 14
2.2.3 Integration of data 15
2.2.4 Imputation and estimation of data 16
2.3 API Access 17
2.4 Security 18
3. Evaluation 19
3.1 The Netherlands 19
3.2 Slovakia 22
3.3 Italy 23
3.4 Poland 23
3.5 Portugal 24
3.6 Bulgaria 25
3.7 Greece 25
4. Software 26
4.1 System design 26
4.2 Backend 26
4.3 Frontend 27
4.4 Security and testing 27
5. Conclusions 29
References 30
Annexes 31
The Netherlands – data analysis 31
Building user-friendly and flexible solutions in R 34

Executive summary

The deliverable J4 is a technical report containing the results of conceptual work of the workpackage J *Innovative Tourism Statistics*, which are the basis for preparation of a Tourism Integration and Monitoring System (TIMS) prototype. The main purpose of the report is to provide a description of the concept of the prototype with particular emphasis on the system components.
The report consists of five chapters.

The first chapter describes the problems arising from differences in statistical production processes in individual countries and the solution in the form of a prototype concept of a comprehensive system for collecting and processing data for statistics in the field of tourism. The chapter focuses on presenting the general assumptions of generic statistical models describing statistical production processes and relationships between data. An analysis of the requirements for the presented system has been also carried out.

The second chapter describes the prototype of the system, focusing on its main components. This chapter discusses the following modules: data collection, integration, imputation, estimation and system management module. The issues of data access as well as sharing of collected information via the application program interface (API) are also discussed. Security issues for processed data have also been described.

The third chapter describes the evaluation of micro-services that potentially could be part of the prototype. Advantages and disadvantages of IT solutions and their implementation in partner countries have been also described.
A list of tools and the planned way to implement them as part of the developed prototype system, broken down into system design, backend, frontend and security and testing, is provided in **the fourth chapter**.
The last chapter of the report contains a summary of conceptual work implemented by the WPJ team, conclusions and recommendations.

Glossary

API (Application Programming Interface) is a way, understood as a strictly defined set of rules and their descriptions, in which computer programs communicate with each other.

GSBPM (Generic Statistical Business Process Model) describes and defines the set of business processes needed to produce official statistics.

GSIM (Generic Statistical Information Model) is a reference framework of internationally agreed definitions, attributes and relationships that describe the pieces of information that are used in the production of official statistics (information objects). This framework enables generic descriptions of the definition, management and use of data and metadata throughout the statistical production process.

HTML (HyperText Markup Language) is a hypertext markup language that allows one to describe the structure of information contained within a web page, giving meaning to individual fragments of text – forming hyperlinks, paragraphs, headers and lists - and embedding file objects in the text.

JSON (JavaScript Object Notation) is an open standard file format, and data interchange format, that uses human-readable text to store and transmit data objects consisting of attribute–value pairs and array data types (or any other serializable value).

Metadata is a description of other data.

RSS (Rich Site Summary) is a web feed that allows users and applications to access updates to websites in a standardized, computer-readable format.

XLSX (Office Open XML) is a zipped, XML-based file format developed by Microsoft for representing spreadsheets, charts, presentations and word processing documents.

XML (Extensible Markup Language) is a universal markup language designed to represent different data in a structured way. It is platform-independent, which facilitates the exchange of documents between different systems.

Web scraping (also known as Screen Scraping, Web Data Extraction, Web Harvesting, etc.) is a technique used to extract large amounts of data from websites.

Web service is a service offered by an electronic device to another electronic device, communicating with each other via the World Wide Web.

1. Introduction

Rapidly changing and available new technologies require statistical offices to adapt their IT systems to collect and process data, develop them continuously, as well as to improve the consistency and comparability of results obtained on the basis of data gathered also from unstructured big data sources.

The pilot project of WPJ and its envisioned output have the challenge to provide a prototype (IT-system tool) to monitor changes in the Tourism sector. The potentiality and corresponding feasibility of this prototype (set of system components) are based on the experiences gathered and tested across the eight countries participating in the WPJ, and more importantly, it relies greatly on the assistance of an IT- platform and its well-performing components.

The current COVID-19 pandemic crisis has hit all economic activities. The tourism industry, in particular, has been impacted hard. Tourism interacts with many other economic activities such as transport, retails industry, sport & culture and communications, causing many indirect negative effects throughout the economy.

This crisis has shown how important it is to have a monitoring system in place that is able to detect and visualize these changes rapidly in the EU. Before the COVID-19 crisis, the motivation for having such a monitoring system was (amongst others) the threat of mass tourism in famous cities. Now, such a system could be used to measure how fast tourism is starting up again when many countries start to ease their travel restrictions.

Currently there is a need for quicker, more disaggregated and up-to-date information for stakeholders in the Tourism industry, the Government and City Boards. Thus, last challenge is to provide Flash estimates that would enable EU member states to monitor the changes and impacts of reopening their economies to local and foreign visitors. This requires a suitable IT-system assembled with ready-to-use system components, as described in this document.

1.1 Technical review

Across the European Union statistical organizations conducting research in the field of tourism undertake similar activities, although each country uses different processes to describe individual phenomena. Processes associated with the acquisition, processing and publication of the results, however, may vary depending on the country and institutions. For example in many countries establishments with less than ten bed places are not covered by surveys. Additionally, different thresholds are used in specific countries to collect information on establishments. The frequency of data collection is not very diverse and usually data are collected on a monthly basis, while differences often occur in the observed population of establishments as well as methods of data collection. All this hinders communication, exchange of tools and methods in the framework of cooperation between statistical organizations and comparison of obtained results. In addition, the use of big data

sources in this process is an additional challenge in the area of methodology and applications of modern technologies.

One of the solutions to the problems mentioned above can be the application of the Generic Statistical Business Process Model (GSBPM), which describes all stages of statistical production. It will allow the unification of diverse statistical surveys carried out in individual countries by defining, describing and mapping processes in a standardized manner. In addition, it will enable the inclusion of uniformly described data from non-statistical sources, which until now have had an unstructured form, in official data.

The use of GSBPM will also ensure comparability of data held by European Union statistical institutions and will allow the creation of a quality control framework. Thanks to this, it will be possible to exchange solutions and methods developed in each country and to create one system for all tourism statistics.

GSBPM divides process components into individual phases (blue colour) and sub-processes (pink colour), which are presented in Figure 1.

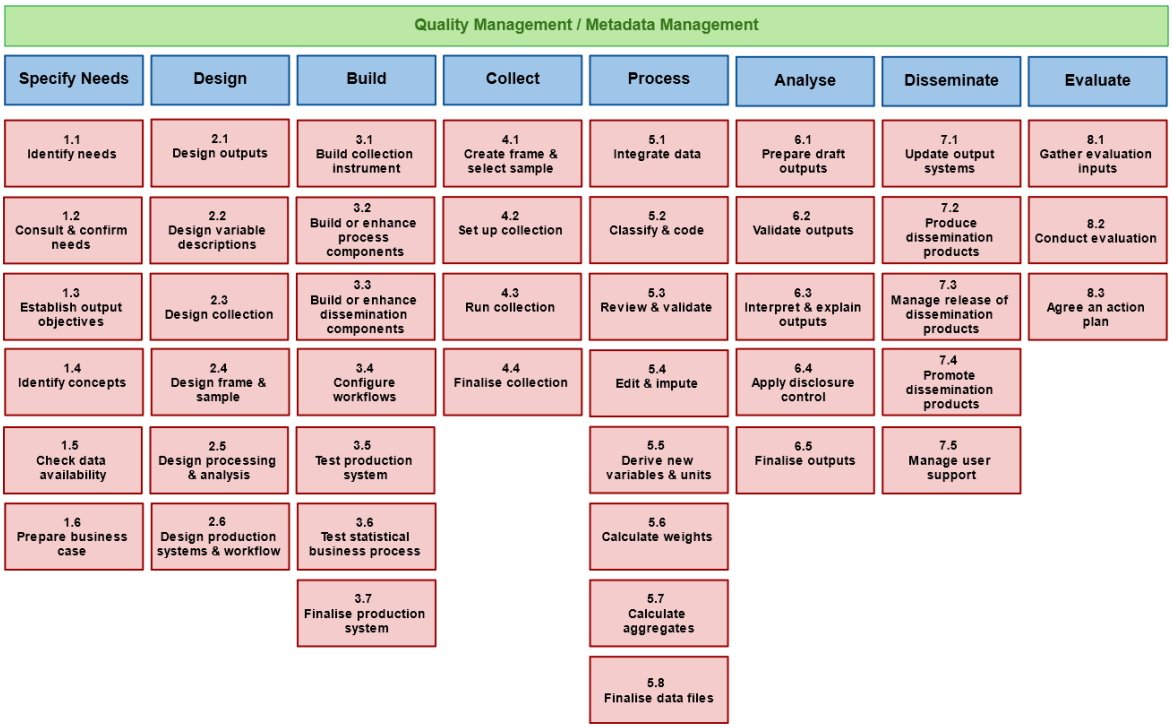

Figure 1. Structure of the GSBPM model divided by phases
(Source: https://statswiki.unece.org/display/GSBPM/IV.+Levels+1+and+2+of+the+GSBPM)

However, it should be noted that all statistical production sub-processes in the GSBPM model are not strictly ordered. The model itself is not a linear model, it does not require the use of all sub-processes, does not define the order of transitions between them and even assumes the possibility of iterative loops between sub-processes.

Therefore, it is necessary to organize information in the GSBPM statistical process. For this purpose, the first reference framework for statistical information, called the Generic Statistical Information

Model (GSIM), has been developed and approved at international level. This framework plays an important role in the modernization, improvement and adaptation of standards and production related to official statistics at both national and international level. GSIM has been designed to allow an innovative approach to the production of statistics as far as possible. It specifies general descriptions for defining, managing data and metadata throughout the entire statistics production process. It provides a set of standardized, consistently described information objects that are input and output data for designing and producing statistics. It also describes the relationships between entities involved in the production of statistics and can be used to guide the development and application of consistent standards or implementation specifications. It is one of the foundations for the modernization of official statistics and the departure from thematic silos, as well as a key element of the strategic vision prepared by the High Level Group for the Modernization of Production and Statistical Services (HLG) and approved by the Conference of European Statisticians.

GSIM does not however, include information objects related to business functions in an organization, such as human resources, finances or legal functions, except in cases where this information is used directly to create statistics. Instead, it contains objects that specify real world information - *Information Objects*. Examples of such type of objects include data and metadata (such as classifications), as well as the rules and parameters needed to run production processes (for example, data editing rules). GSIM identifies about 110 information objects, which are grouped into four top-level groups and a Base Group, which be seen in Figure 2.

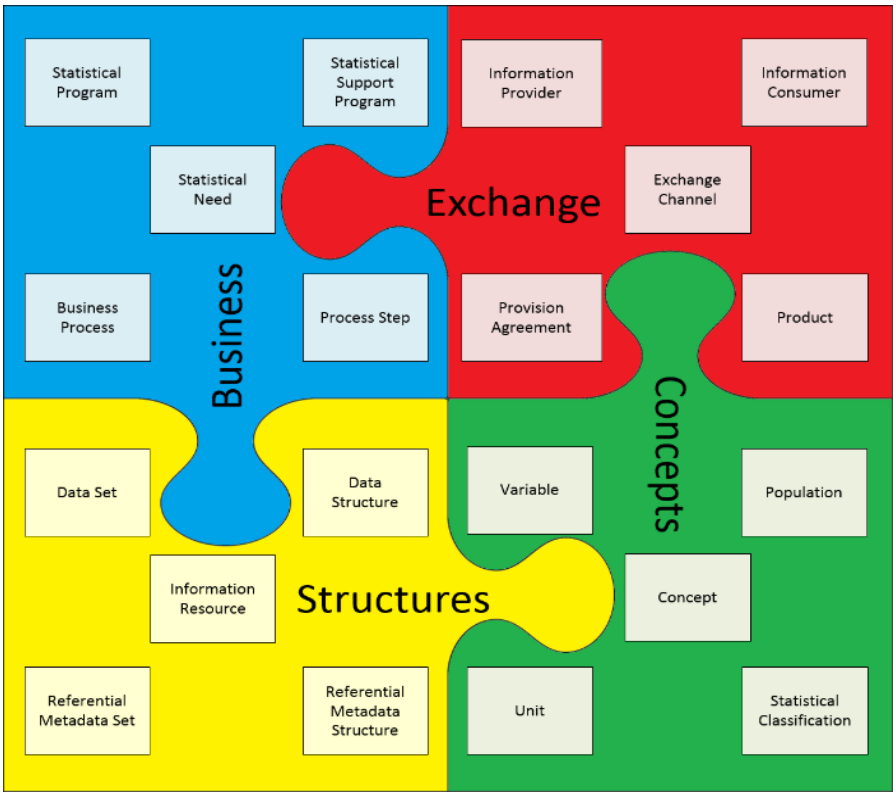

Figure 2. Top level groups of the GSIM model with information objects
(Source: https://statswiki.unece.org/display/gsim/Generic+Statistical+Information+Model)

The four top-level groups are:

- **The Business Group**
  used to capture projects and plans of statistical programs and processes undertaken to provide these programs. This includes identifying the statistical need and business processes that make up the statistical program and their assessment.
- **The Exchange Group**
  used to catalogue incoming and outgoing information from a statistical organization through Exchange channels. Includes objects describing collection and disseminating information.
- **The Concept Group**
  used to define the meaning of data, providing an understanding of what data are measuring.
- **The Structure Group**
  used to describe and define terms related to information and its structure.

GSIM is therefore a complementary model providing the structure of an information object supporting all statistical production processes (such as those described in GSBPM) and, at the same time, providing the agreed names of information objects, defining them, specifying their essential properties and indicating their relationships with other information objects. Nevertheless, the GSIM model does not make assumptions about the methods or technologies used to implement the model.

A detailed description of the models used to build the WPJ tourism statistics system was developed earlier by the workpackage F of the ESSnet Big Data II project and presented on the example of the BREAL system in the deliverable F1 entitled *BREAL: Big Data Reference Architecture and Layers. Business Layer*. The document also contains a description of layers, actors and data life cycle, which also served as the basis for the prototype system for tourism statistics described in this document.

1.2 Design requirements

Due to continuous changes, not only technological ones, but also socio-economic, cultural or changes in the research methodology and availability of data sources, the designed system, should have features that will allow its future free use by various IT tools and allow its further development based on new sources that are currently unknown or do not exist at the moment.

Therefore, during the work on the project, an analysis of the system requirements was carried out and the following assumptions were defined for an IT tool that would continuously provide integrated access to comprehensive information necessary from the point of view of tourism statistics. These are the following assumptions:

1. Building a system based mainly on open-source solutions.
2. Use of already created tools in the form of micro-services.
3. Collection and integration of data from various sources and in various formats.
4. Built-in module for downloading data from web portals.
5. Cleaning and normalization of collected data.

6. Integration and imputation of missing data using machine learning algorithms (artificial intelligence).
7. The ability to customize algorithms for individual methods of calculations.
8. An extensive audit mechanism enabling tracking of changes and operations performed in the system.
9. Access to the collected data through the so-called mechanisms web-services and API for statistical organizations.
10. Keeping statistical confidentiality, provisions on the protection of personal data and protection of copyrights.

After analysing the needs, WPJ decided to build a system for tourism statistics based on the distribution of program functionality into independent, interchangeable modules, so that each contains everything necessary to perform only one aspect of the desired functionality. In this way the following parts have been distinguished:

1. Data collection module that retrieves data from relevant network or base sources.
2. Metadata module containing a description of individual sources and variables stored in the system.
3. Data normalization module ensuring uniformity and comparability of data and conversions.
4. Optimization and imputation module.
5. A management module coordinating the operation of all other modules, in particular responsible for the timely execution of individual processes.

Each of the above modules will include a number of related micro-services, i.e. many components implementing strictly defined, relatively small fragments of applications, which will allow the use of tools created and currently used in official statistics.

2. Prototype description

2.1 Overview

In response to contemporary needs and trends, as well as prospects for further development, the TIMS will be designed using currently used modern information technologies. Its construction will use programming languages and libraries available as free/open software. The entire system infrastructure will be prepared on the basis of a set of rules that define communication between computer programs (API) in the REST architecture (client-server). Adoption of such a solution will ensure a high versatility of the tool. In addition, all queries sent by a client will be authenticated, which will protect the system against unauthorized access to data.

The solution being developed fulfils the task of integrating data sources identified in the field of tourism statistics in the system databases. In addition, it aims to support the production process of statistics (including experimental ones), as well as data from both statistical databases (statistical surveys), data made available by external providers and those obtained from big data sources

(e.g. web scraping, social media). The integration of data sources will be supported by their analysis and documentation in a metadata system that allows describing the variables and relating them to each other.

Thanks to the implemented internal mechanisms, the designed TIMS, will allow for pre-processing of data, including cleaning, unification of their structure and saving to databases. At a later stage, it will be possible to analyse the collected data and process it to obtain the final results.

The system will operate based on a number of individual modules, whose number and scope can be successively expanded depending on the needs of users. Dedicated modules (e.g. queries, integration, calculation, etc.) with implemented mechanisms, functionalities and methods of data estimation, will allow for data to be filtered, supplemented (imputation of missing data) as well as calculated and presented in various formats. The output data will be disseminated using existing official statistics systems, as well as via a dedicated API.

Figure 3 presents a use case diagram showing possible user interactions with the system. Solid lines represent user interactions with the system, while dashed lines represent related services that are called automatically.

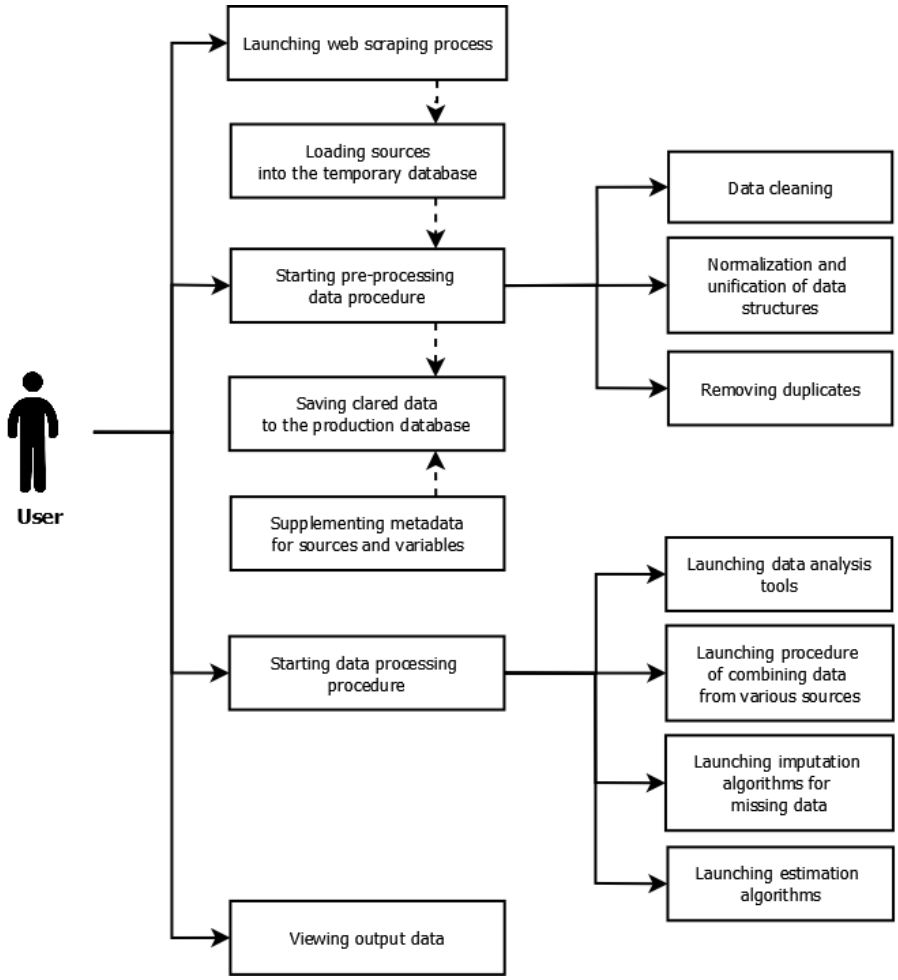

Figure 3. System use case diagram

2.2 Prototype components

An analysis of the needs of the WPJ partners involved in the Big Data II project and an inventory of sources available in the field of tourism statistics showed the need to develop a universal system at the European level tailored to the individual needs of national offices and characterized by scalability. In response to these needs, the prototype uses a modular structure that will facilitate the modification of currently implemented elements in the future and allow for easy expansion with new ones.

As part of the prototype system four elements can be distinguished:

- navigation - involves moving within a module, as well as between modules,
- view - i.e. the issue of displaying user data,
- user actions - primarily working on data, updating them, etc.,
- data binding - a separate data layer allows to make the data independent of the user's view and actions.

The functional diagram of the prototype is shown in Figure 4.

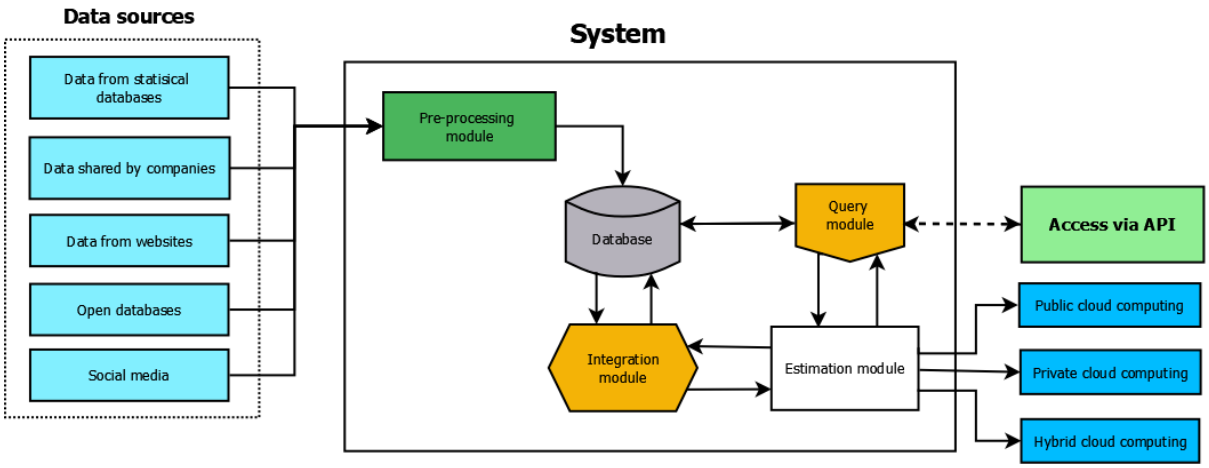

Figure 4. Functional diagram of the prototype

According to the diagram presented above, the designed system will have support for data from various sources such as:

- statistical databases,
- websites downloaded by web scraping,
- open databases provided by external entities,
- social networks (e.g. Facebook, Twitter),
- databases made available by companies in the form of JSON, RSS or web services.

The data will then be cleaned and normalized to a uniform structure and loaded into databases.

The next component of the system is optimization and filling in missing data. Machine learning algorithms will be used for this purpose. Processed data require enormous computing power, therefore, cloud computing will be used for this task:

- public (Amazon AWS, Microsoft Azure, Google Cloud),
- private (OpenStack, Microsoft Hyper-V, VMWare, Oracle VM),
- hybrid, combining selected functionalities from public and private clouds.

Communication with the tourism statistics system will take place using the API which will allow separating the user interface from server operations. It will also allow access to applets and scripts for the client to perform specific operations on the data.

2.2.1 Data gathering

The data collection module should enable loading data from various sources, including websites, relational databases, flat data files, i.e. CSV, Excel, txt. The module will allow to connect to APIs, databases, and then gather data from them based on specific searches or other conditions. Each process of loading data from various sources can be extended by adding additional steps that allow to save the collected data to a specific location in the selected database. Common elements from different sources can therefore be saved in the target database, even if the naming conventions and/or data formats of these data points may vary depending on the source.

One of the key elements of the system is the ability to download data from web portals. WPJ has developed a Visual Modeler (VM) micro-service that will be a part of the data collection module. The tool was created based on the open-source solutions available on the market, which ensures high portability for various platforms.

The VM (see Figure 5) has a graphical interface and can be used to download data from any website. In addition, this tool can be used for data processing operations (analysing, conducting calculations, etc.). The use of the application does not require advanced programming knowledge from the user.

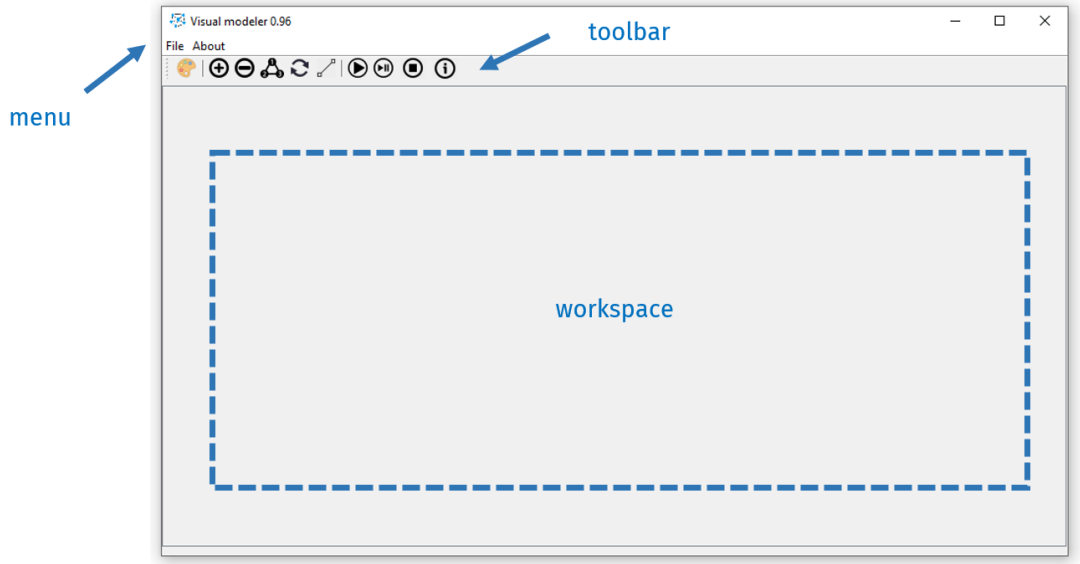

Figure 5. Visual Modeler micro-service

Special functional components built into the palette and buttons on the taskbar are used to create the process (program) in the application (see Figure 6).

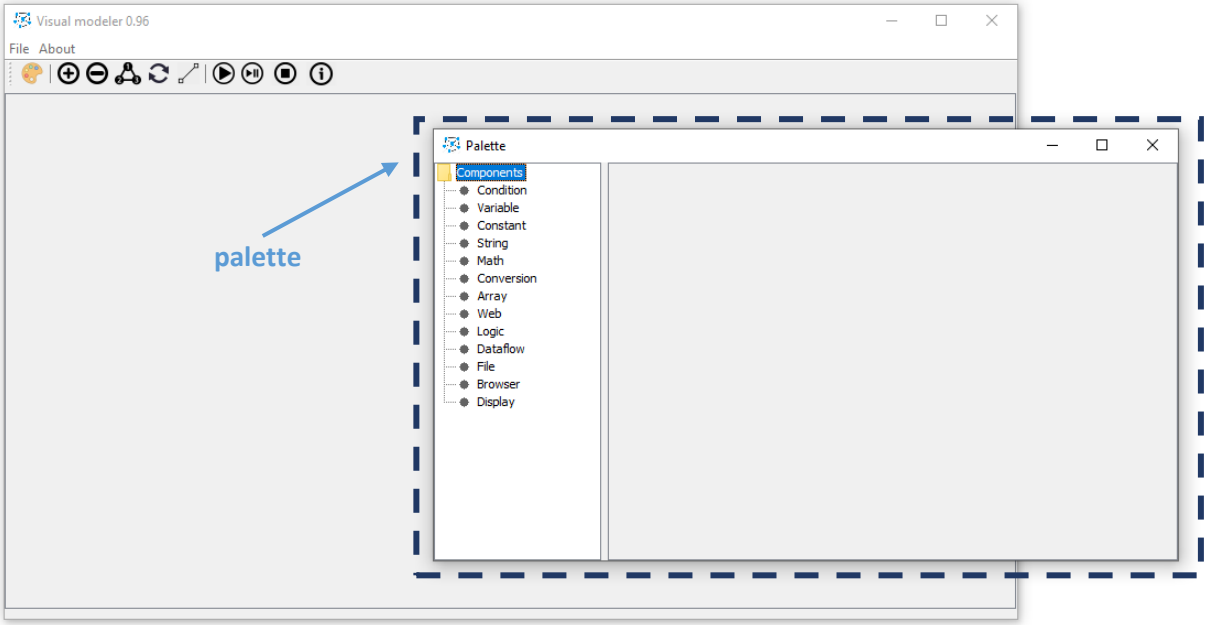

Figure 6. Components palette

Within the palette, groups of the following components are available:

- Condition - a group of components containing mathematical conditions for comparing quantities (equal, greater, greater or equal, smaller, smaller or equal, different),
- Variable - individual components of the group are responsible for downloading and setting variables,
- Constant - a group of components defining the types of variables and the operations that can be performed on them,
- String - components for performing actions on text elements are located in this group;
- Math - a group includes basic mathematical operations and functions,
- Conversion - components in this group are responsible for converting one type of variables used in the implemented process into another,
- Array - the group includes activities on tables,
- Web - a group used to build a process related to websites, also containing components responsible for navigating the user on the website,
- Logic - a group of basic logic operations components,
- Dataflow - components contained here refer to iterative operations such as loops, jumps and relate to activities that are to be performed once during one session of the process,
- File - with the help of the built-in options the user can perform operations on directories and files,

- Browser - a group of components dedicated to activities performed in a web browser on a website,
- Display - a group of components used for visualization.

The tool allows to create a web scraping process directly in the application using the components presented above, load data from external APIs or scripts created in other languages (javascript, R). Each prepared process can be automated using tools available in the system (e.g. in Windows - Task Scheduler), and the built-in error service will inform if the process cannot be performed. An additional tool description can be found in Deliverable J2 *Interim technical report showing the preliminary results and a general description of the methods used*.

The construction of the Visual Modeler micro-service as an integral part of the system will allow easy access to new types of sources for all statistical organizations, regardless of their programming knowledge. It will also ensure compliance with good practices related to the web scraping process itself. In addition, the advantage of the micro-service will be to prepare for all statistical organizations complete processes of downloading data from web portals and their recording in a uniform structure, which will facilitate the comparison of data for individual tourism-related phenomena.

2.2.2 Pre-processing

The pre-processing module should provide a mechanism for data pre-processing, in particular as regards the unification of data types, e.g. conversion of numbers from 16-bit to 32-bit, etc. Target data types should be specified at the design stage of the main database where all information will be stored. Due to the fact that different sources save different information in different ways (e.g. true can be saved in different systems as: 1, true, "TRUE", TRUE, HIGH), unification is necessary for efficient integration and subsequent search data.

This module is a bridge between the *outside world* and the internal representation of stored information in the form of a database. The module should provide functionality that allows to load data from an external source, but in its basic form should contain an abstract implementation of the necessary functions implemented in specialized modules. The mechanism should only define the general form of objects, leaving details to derived classes. It does not specify the exact types and does not provide implementations of methods. Due to the definition of the abstract base structure, the system may be extended in the future with another micro-services for data types that will appear after the system has been implemented, without the need to expand or modify it. This is an important advantage of the system, which will be extensible along the lines of the so-called plug-ins.

The WPJ has prepared a micro-service based on the HERE Maps API, which allows geocoding address data. It supports files in various input formats, provides automatic correction of address features (such as typos or abbreviations) and quality control of generated coordinates. The micro-service will provide a common structure for identifying address characteristics for all entities using the system.

2.2.3 Integration of data

The use of data from various sources or systems made available by external providers in the designed system is also associated with the need to supplement the obtained data and optimize them. The lack of some data or selected items in the data and the impossibility of obtaining them only from one source should be taken into account. This may result from the needs of users of specific systems supplying the TIMS databases. The interests of hotels owners may focus on obtaining and collecting information on the number of guests and the length of their stay and not information related to the country of residence or food expenses. These, however, will be crucial when creating statistics for various areas of tourism.

Therefore, one of the most important modules of the designed TIMS is the data integration module, i.e. a module that performs the tasks of cleaning data, removing contradictions, combining information from various sources, often incompatible. Therefore, the module in question must be meticulously designed. The quality of information collected in the database depends on the quality of its functioning.

Considering the requirements, i.e. entering data into the database in the form of a unified field structure and access to only partial information from various media, the most optimal solution is to choose a hybrid platform combining different techniques into one coherent whole. The hybrid technology should include the following technologies:

- data integrator from relational databases (Oracle, MS SQL, MySQL and others),
- flat data source integrator (CSV, txt and other files),
- data integrator from network sources (HTML, JSON, XML files).

The integrator's function is to process data from the input form, e.g. HTML, to the form of relational database fields in the system. The integrator should be a filter that converts one data stream into another, the appropriate implementations of the filter function should be saved as universal libraries. It should be assumed that the data after applying the filter will be incomplete, due to the limitations of the source. It is completely natural and in line with the above assumptions. Fields that cannot be saved in the database due to missing input data should contain a special value indicating that they can, if possible, be replaced by appropriate values.

Data linkage micro-service developed by WPJ package enables to check if a given tourist accommodation establishment from the portals is also present in the survey frame. It supports two approaches to linking of data: deterministic (or rule-based) record linkage and probabilistic (or fuzzy) record linkage. In deterministic matching, an identifier or a group of identifiers across databases are compared. If it is not available then a group of identifiers can be used e.g. postal code and address.

The micro-service for web scraping described above contains several identifiers that can be used for the linkage:

- name of an accommodation establishment,
- address,

- postal code,
- longitude and latitude.

Probabilistic (or fuzzy) record linkage uses geolocation to match two establishments. The details about those approaches can be found in the J3 deliverable “Methodological Framework Report”.

2.2.4 Imputation and estimation of data

Missing data may be crucial during processing and searching for certain dependencies, hence there is a need to supplement them, if possible. For this purpose, appropriate imputation mechanisms should be identified and developed in the system, and relevant, efficient computational tools should be prepared.

At the initial stage of the imputation process, missing data should be searched in sources of a similar nature, e.g. relational databases, CSV files, etc. In this case, it is not required to use intelligent solutions - ordinary data integration mechanisms are sufficient, such as appropriate SQL queries, hashing tables or dictionaries. If this approach does not bring the desired results, then the latest machine learning algorithms can be used with the appropriate programming libraries, i.e. TensorFlow, Keras.

Programming libraries should contain currently used methods of data imputation including single and multiple imputation. Due to the fact that the range of methods used in the European Union countries is very different, methods ranging from the simplest ones such as Mean imputation, Hot deck imputation, Cold deck imputation to, for example, Maximum Likelihood Estimation (Dempster, Laird, Rubin (1977)), Deep Learning (Camino R.D., Hammerschmidt C.A., State R. (2019)) or Multivariate Imputation by Chained Equation (MICE) (Buuren, Groothuis-Oudshoorn (2011)), should be considered. The implementation of these methods can be found in libraries of many programming languages such as R and Python.

After the data imputation process, actions that are specific to official statistics only can be undertaken. They include, among others:

- calculation of weights,
- calibration of weights (linear, raking, logit, truancated),
- data aggregation (Horvitz-Thompson estimator, GREG estimator, Calibration estimator, etc.),
- estimating precision of results (bootstraping and resampling),
- seasonal adjustments (TRAMO-SEATS and X-13ARIMA-SEATS),
- benchmarking (Denton’s additive and multiplicative approach, etc.).

The system should enable these procedures or easily integrate software dedicated to these procedures.

As part of the work, two micro-services for Flash Estimates and Temporal Disaggregation were built in R.

Depending on the size of the data sets, the process of data imputation and estimation can be time consuming. If local machines in the form of data servers do not provide adequate computing power,

then the resources provided by public (commercial) cloud computing from known suppliers can be used. The alternative is to build a cloud computing. However, this involves the need to have an appropriate server room and appropriate strong hardware infrastructure.

In the first case, cloud configuration is much simpler and there is no need to have own hardware. On the other hand, however, the service provider charges fees for each processor calculation cycle, disk space and generated network traffic. Costs can be high if there is a significant demand for power and data space. Developing one's own computing cluster will significantly lower the costs of its operation, even though the power that commercial clouds offer will not be available. The fee will only include the cost of electricity consumed by the servers, whereas disk space, network traffic, and processor load will not generate additional expenses.

If none of the above solutions are acceptable, then a reasonable compromise between high computing performance and the costs that need to be incurred is to use a hybrid cloud. It is a combination of public cloud elements with private cloud elements. Tasks that are not possible to perform on a private cloud due to low performance can be outsourced on a public cloud, while those less demanding - on a private cloud.

2.3 API Access

At the stage of developing the system prototype, it was assumed that it should allow quick and easy search of information collected in the database. Access to data in the form of a dedicated application does not seem to be a good solution here. Such an application is usually closed and available only for the selected operating system, and in addition, if the data search mechanism is changed or expanded, the application must be rewritten.

The solution proposed by WPJ consists in preparing an appropriate API available for any communication system that can send properly formatted text strings. The most popular data formats are JSON, SOAP supported by WSDL and REST architectures. Due to the openness of the formats, the interface can be written in any programming language, and any modifications are easy to implement. In this way, by implementing contemporary design paradigms, application logic, definition of appearance of individual user screens and manipulations will be separated.

Separating the user interface from data storage will not only improve the portability of this interface to multiple platforms, but also allow the development of various components independently of each other. By using the client-server architecture, the state of the user's session will not be stored on server side, and the response the user will receive from the API will be defined as cacheable or non-cacheable. This is important for data that quickly becomes outdated and for data that updates relatively rarely. The decomposition of the system will allow for a number of stateless services and the introduction of an abstraction layer for data storage. This will ensure an appropriate level of system scalability necessary as the amount of processed data increases.

2.4 Security

Due to the use of personal data of users and entities in the system, it is necessary to pay special attention to ensuring an appropriate level of security of processed data. For this purpose complementary organizational, technical and physical safeguards will be used. The TIMS will meet the protection requirements through the built-in basic security attributes such as:

- authenticity,
- availability,
- integrity,
- non-repudiation,
- confidentiality,
- accountability.

The approach to system security will be equivalent to all principles used in official statistics. Applicable ordinances, security policies, including, as the primary, United Nations Fundamental Principles of Official Statistics, procedures and obligations arising therefrom will fully apply to ensure an adequate level of security.

All the modules of the official statistics security system will be used to secure the TIMS:

- Firewall,
- IPsec VPN,
- Intrusion Prevention System (IPS),
- Application Control,
- URL Filtering,
- Anti-Bot,
- Antivirus,
- Identity Awareness.

The firewall system used integrates access control mechanisms, authentication, content inspection, monitoring, intrusion detection, event logging, and provides functionality that allows to create reports. Security will also be enhanced by the appropriate configuration of a firewall area that does not belong to either the internal network or the external network (DMZ zone) and the allowed flow of information accepted by the set rules. The use of network devices that increase the security of IPS computer networks will be used to detect and prevent intrusions by blocking attacks in real time. The key element here is the signature database built as new attacks appear and is updated frequently. System security will be significantly strengthened by processes such as monitoring of protected zones, monitoring of IT systems, vulnerability monitoring, antivirus security, distribution of security patches for servers and workstations, and proper handling of security incidents.

In addition, the system will be prepared in the form of a web application and access to data will be properly secured using cryptosystems, e.g. Pailier. Paillier is a type of keypair-based cryptography. This means each user gets a public and a private key, and messages encrypted with their public key can

only be decrypted with their private key. Unlike many other keypair cryptosystems, Paillier provides "additive homomorphism." This means that messages can be added together while they are encrypted, and they will decrypt correctly.

3. Evaluation

As part of the work, methods to implement new data sources to create experimental statistics on tourism were developed. For this purpose, IT solutions were prepared to implement the methods of:

- gathering data from web portals,
- geocoding of data from web portals and survey frames,
- linkage and Imputation,
- flash estimates of occupancy of tourist accommodation establishments,
- disaggregation of tourists travel and expenses data

in all partner countries.

Each WPJ partner carried out micro-service tests in their country and prepared evaluation of strengths and weaknesses of the solutions. Due to the specificity of available data sources, software used, tourism potential of the country, the uniform structure for case studies was not imposed for project partners. Additionally, the COVID-19 pandemic resulted in delays in testing micro-services. Therefore, the presented material is more comprehensible for some countries.

3.1 The Netherlands

The script was run with data from survey and web scraping data from Hotels.com. A linear model option was used. The model has run for several large cities and NUTS2-regions. Some forecasts looked plausible. In some cases, however, the number of guests or nights in the forecast was negative. A possible explanation for these negative values could be that the model does not work very well when the data is very far away from the training data.

ARIMAX to model the season effects requires a full year of web scraping data and therefore cannot be used yet (data available since July 2019).

Adapting this script for the Netherlands was easy. The Netherlands file paths were only put in the script. The output of the script gives daily values for some variables, but not all. Preparing the data files with survey data took a small amount of time (could be automated in the future, using our 'open data' web service). Therefore, the script can be run by anyone with basic R skills.

A small technical issue occurred: the script searches for a region in the names of the web scraping files. It also matches part of the file. This created an error because both "Amsterdam South, Amsterdam, Nederland" and "Amsterdam, Nederland" occurred. Both matched the text "Amsterdam, Nederland". To solve this, one of the files was removed.

According to the Netherlands, this script can be run by anyone with basic R skills. Some documentation or a short manual could help someone who is using the script for the first time. It is assumed that all required R-packages and updates are available in the National Statistical Offices.

The results obtained using the created micro-services have been presented in Annex: The Netherlands - data analysis.

Table: Evaluation scripts

Country	The Netherlands
No. Testers	2
R version	3.6.2 en 3.5.2
RStudio version	1.2.5033

				MODULE 1 (tester 1) "Hotels com.r" (v3)				MODULE 2 (tester 2) "Flash Accommodation Establishments.R" (v2)		MODULE 3 (tester 2) "Temporal Disaggregation.R" (v1)	
Feature to be assessed	**Feature information**	**Scale**	**Score**	**1a) data analysis and generating monthly / daily stats and establishments data base**	**Score**	**1b) data linkage and imputation**	**Score**		**Score**		**Score**
Data configuration control (default)	**Does script start correctly (with demo data)?**	**best situation – starts correctly** **worst situation – does not start**	**5** **1**			5			5		5
Data configuration control	**Did script require changes to be run?**	**best situation – minimum changes** **worst situation – changes in varios lines**	**5** **1**	Extra step is added to eliminate foreign accommodations. This step uses the variables "lat" and "lon" scraped from hotels.com. Analysis of prices is based on three options: 1) "next day", 2) "last thursday of month" and 3) "next weekend". The output of 1) and 3) are similar. Option 1) is used for the rest of the evaluation. Next to trimming, an extra step to eliminate outliers is included, i.e. "delete if price > 10 thousand euro's". Tunning the trimming parameters. The default parameters seem to work well for the cases 1 and 3). The aggregations to daily and monthly data require little changes, only those regexs to eliminate strings to standardize the names of the locations. NB. This script generates Flash estimates input.		Merging the survey and the web scraped data is executed using geo spatial distances. This task uses Haversine distances and a Distance matrix to link the survey- with the scraped data. The imputation of a set of accommodation features is performed based on Recursive Partitioning and Regression Trees. The evaluation of this sub-module is undergoing.	3	Changes of working paths and recovering a typo	4	Changes of working paths only	4
Integratibility and compatibility	**Is the script implementation country specific?**	**best situation – country independent** **worst situation – country dependent**	**5** **1**	At the beginning we had problems with the libraries which were not standard available in our IT-services. But, this is solved now.					5		5
Tailorability	**how much time is needed to adjust code and prepare inputs to your case?**	**days**	**[0,Inf)**	Most of time is used to check whether the matching has been performed correctly.	2 ≤ t ≤ 4	An OC summary and plot are provided as tool to monitor the quality of the imputation.			t ≤ 1		t ≤ 1
Response times	**how quickly you can produce outputs (if the inputs are prepared and code adjustments**	**best situation – you can run whole code without any intervention** **worst – you check every line each time you run it**	**5** **1**	The first part of the script reads 2.6 x 10^6 records in a few minutes.		The Distance Matrix is efficiently computed			5		5
Domain of tool	**To what extend it produces an auxiliary information needed to assess the quality of outputs or to adjust the parameters of the implemented methods .**	**best situation – you can review many intermediate statistics and plots which give you data insights** **worst – you can see just the final output**	**5** **1**	The built Input/Process/Output indicators are well-chosen and self-contained. Script contains indicators on coverage, "Number of day of scraping". This can be considered as a performance measure of the scraping-process and also about the reliabiliy and completeness of the webscraped data.		5		5	5		5
Skills base and user capability	**To what extend a person (e.g. R-newbie) can use that script?**	**best situation – everybody can do it without a deep introduction** **worst – it requires a person advanced in R and a deep introduction to what that script does)**	**5** **1**	Intermediary knowledge of R is needed.		3 Intermediary knowledge of R is needed.		Basic knowledge of R is needed.	3	Basic knowledge of R is needed.	5
Additional comments or remarks											
Integratibility and compatibility	**Use R package "eurostat" to download open data from the Eurostat database. It contains scripts targeting data search, download, manipulation and visualization of among others NUTS SpatialPolygonsDataframes.**	**best situation – package compatible** **worst situation – package incompatible**	**5** **1**								
Integratibility and compatibility	**The geo tool and its output are useful to check the completeness of the variables "lat" and "lon". We use QGIS to eliminate the foreign companies and import back the resulting csv file.**	**best situation – output compatible with other open source tools** **worst situation – ouput incompatible with other open source tools**	**5** **1**								

Table 1. Summarised evaluation of the micro-services

3.2 Slovakia

Testing started with R code Hotels_com.R, which analyses scraped data, prepares outputs for temporal disaggregation and flash estimates, creates database of unique accommodation establishments and joins it with the statistical frame of accommodations and imputes some of the missing characteristics.

Each single line of the code has been reviewed and raised some adjustments, comments and efficiency improvements have been raised. As the code is mainly created with the Polish data, some national arrangements were necessary.

First part of the script deals with daily data scraped from the hotels.com page and creates some histograms, time series and statistics on mean and median prices, with respect to the type of the accommodation. An important output is the database of unique accommodations scraped during the whole period, which can be further analysed and used for updating of the population frame. This part of the code is very standardized, since the data scraped from hotels.com have the same structure for each of the partners. The only necessity is to set the working directory and the folder with scraped data and run the whole part of the script. The outputs are created and saved straightaway. The time needed for this regular operation is about 2 minutes. In the case of Slovakia, however, some further adjustments were crucial, especially in terms of regional variables, which are missing in Slovakia's scraped data (scraping for the whole country is done at once) and need to be joined using postal codes and municipality names. Furthermore, adjustment of accommodation and municipality names and addresses into a common standard is also performed.

The second part prepares input data for temporal disaggregation and flash estimates. This part have been revised and it also works properly, similar to the first part. The outputs are saved into the directory. The time needed to perform this action is below 1 minute and it should work for any data scraped from hotels.com using the Polish tool.

These two parts do not require any special R knowledge and can be used by almost any statistician.

The third part deals with linkage of database of scraped data and statistical frame. However, this part is strongly country-specific and cannot be run on different datasets. Taking into account that the national frames can vary a lot, extensive adjustments to the code would be necessary. Moreover, the code cannot be run even with the training Polish data provided, as there are some differences. From the point of view of Slovakia, the code lacks deeper analysis of distances between coordinates, does not makes use of accommodation names and/or addresses and does not deal with string standardization, which makes it useful only to the extent of an example of using a threshold rule for joining with GPS coordinates and only with Polish data. A specific R code for joining of scraped data and accommodation frame in Slovakian national conditions are in the process of being made.

The last part is devoted to the imputation of some missing data in joined database. There are some very nice ideas, e.g. use of decision trees along with ROC curves, but still it is country specific and needs further adjustments to be useful to other partners. These ideas will definitely be used in Slovakia's approach.

These two parts cannot be used by neither R newbies nor R lower intermediates. Adjustments needed for national conditions can take considerable amount of time. On the other hand, taking into account

that the data can vary a lot between partners, it would be very difficult to make a single unified script that could work for everybody. Moreover, R skilled persons can make use of the ideas presented in the code and adapt them in national conditions.

3.3 Italy

As regards Hotel.com micro-service, it runs fine without any issue and produces .csv file containing all accommodations according to the request. Some adjustments have been necessary to customize software for Italian destinations and to overcome the differences in classification of property types with respect to the Polish case. Default browser settings had to be changed to avoid the confirmation for saving each separate output file. At the principle, micro-service ran on some cities and then for the whole country. In the latter case an incomplete result was obtained, i.e. only an extraction/sample of the entire database.

A similar problem occurs in the case of Booking.com micro-service, as it extracts only the first 40 pages (1000 hotels) from the website, when launched on a large territory as Italy. Even trying to increase the offset beyond the threshold of 1000 does not seem useful, as it always returns the same structures obtained on page 40 (every other page later is the same). Better results were obtained using more selective search criteria: the municipality instead of the province, the neighbourhood instead of the municipality in the case of too many structures and so on. Furthermore, in some cases additional filters must be needed to define subset of results for larger municipalities (e.g. the number of stars of the hotel or the type of accommodation). Consequently, a big initial effort is needed for adding all the specific search parameters in order to cover the entire territory. The preliminary identification and specification of all provinces and municipalities to be searched is essential to be almost sure that some of available accommodation data are not missing in the results. Another noticed issue is the bot detection that requires to give (human) input when the Booking portal ask for it. This can occur in the case of high duration of web scraping. The last (marginal) comment on booking.com script is that it provides only html pages of the search results, so another software is needed to extract the text from html pages, in order to produce input for the following analysis (R programs).

Regarding Geo-localization micro-service, it is very user friendly, simple to use and it produces expected outputs. It has some limits about particular keywords, for example the term "Viale" is not recognized by the system but it is frequently used in Italian addresses. Comparison of the results with other geospatial system like ArcGis, Atlas, Egon is still in progress in Italy.

As concerns Visual Modeler micro-service, it is an excellent work tool that can be used in organizations where not many skilled resources for developing web scraping software are available. As a visual tool, it is immediately understandable and easy to use. It runs fine on the selected portals that was given as example.

3.4 Poland

The micro-service for Flash Estimation works properly. No major implementation problems occurred in our country. The received code required minor syntax corrections to adapt to the batch files generated in our country regarding accommodation statistics. A great facilitation in the implementation of the code in our country is the fact that it uses files generated by another micro-service that calculates statistics of prices obtained from web scraping. The structure of this file is the same in every country, which means that part of the code did not require any interference.

It should be taken into account that possible modifications to the code can be made after loading the data. It was the preparation of the obligatory data that required more attention. In general, the time needed to prepare the script and data was slightly more than one day, i.e. about 10 working hours.

The correct preparation of data is key here because after loading them the program generated results in a few minutes. In addition, the program generates statistics that allow to assess the quality of results, which significantly facilitates the decision on the further steps related to their acceptance or rejection. As a result of the script implementation, information on which variables were used in the forecast are obtained. Therefore, sense of variable selection can be assessed.

On the other hand, the script offers only two types of models, i.e. linear models and the ARIMAX model, while it is not possible to choose logarithmic transformation that allows estimation with non-linear models. It would also be a great help for the script to automatically generate charts of real data and those from the model, as well as a graph of model residuals. Alternatively, extending the model assessment capabilities to include econometric residual tests would be an additional tool to assess the correctness of results.

The Temporal Disaggregation micro-service worked correctly with different sets of parameters. It generates daily values aggregated to monthly ones, but the result of the model is not always satisfactory. Options for narrowing of time series manually by entering the start and end dates are available, as well as options for automatic trimming of time series, which significantly saves batch preparation time.

The micro-service allows to choose one of the two most commonly used benchmarking methods. It works on batch files prepared for the needs of the script for flash estimates of the accommodation establishments, so it did not require additional work on batch files. The code itself do not need to be changed to suit local needs.

In general, the script can be executed in its entirety, providing only paths to the working folders. In addition to the disaggregated time series, only information about the R2 square of the LASSO model is available. There are no other statistics and charts allowing to assess the selection of explanatory variables and the quality of the model. The script is able to be operated by a person with little knowledge of R after a few minutes of instruction.

3.5 Portugal

The micro-service used since October 2019 for web scrapping hotels.com website, does not collect latitude and longitude, for this reason this report refers only to “data analysis and generating monthly/daily stats and establishments database”.

It runs smoothly after some minor adjustments and produces the expected output files. Once the inputs are prepared and code adjustments are done the script runs without intervention. It runs very fast even though the input files are very large (the data scraped from hotels.com concerns 308 municipalities and cover both Portugal mainland and Islands Azores and Madeira).

The latest version of the Linkage & Imputation micro-service is complicated and a potential user must have some domain knowledge and some proficiency in R language in order to use it, hence more explicit comments and a more general use of sections would be a very welcome improvement.

When a variable is created (e.g. line 70: accType Stat or line 82: NACE) there should be a new section which explicitly indicated the creation of a new variable and not only a regular / simple comment. This way the user could navigate the document outline more easily and understand what it is being executed along the script. In R language the same result can be produced in a multitude of ways so an inexperienced R user might not be familiar with the solution implemented in the script. The script already has a significant number of comments on the code, extending this good practice would be highly beneficial. This does not mean that the code should be simplified, on the contrary, using efficient code written by experienced users and learning from it is a great way to improve one's programming skills.

Due to the number of output files produced by the service it would be useful to have an html RMarkdown report with all the plots included as well summary information on the CSV files. This way, one would not need to open several individual files in order to have a general idea of the results. Nonetheless, the script and the output provide all necessary elements to make a report.

3.6 Bulgaria

The activities on real web scraping started with testing and then implementing the sample micro-service for the portal hotels.com. The tests finished successfully and up to now daily information for the available accommodation places on the territory of Bulgaria from www.hotels.com are being scraped. The data are saved in files and subsequently processed with micro-services prepared for imputation and estimation of data, which also was tested in Bulgarian conditions and now works successfully. Both work properly without any major problems just with small adjustments at national level. They are user friendly and could be used easily by statisticians. The micro-service concerning the linkage of database of scraped data and statistical surveys data is more or less country-specific and needs to be adjusted to the national specificity and circumstances.

3.7 Greece

The micro-service for data gathering for internet portals, after the necessary adjustments for national conditions, runs well and create useful outputs. The integration micro-service uses the scraped data from hotels.com and creates histograms, mean and median prices by type of accommodation. Another output produced by this micro-service is the dataset of accommodation scraped for the period. This specific output is very useful as it could be further analysed, in order to explore the contribution of the scraped data on the improvement of the completeness and consistency of the statistic Frame of Tourist accommodation establishments.

4. Software

4.1 System design

PRODUCT		USE
Enterprise Architect	ENTERPRISE ARCHITECT	Software for modelling the system using UML diagrams.
Archi/Archimate	Archi archimate modelling	Open-source software for modelling using an open and independent language to model corporate architecture.

4.2 Backend

PRODUCT		USE
SQL Server	Microsoft SQL Server	Database management system, supported and distributed by the Microsoft corporation. This is the company's main database product, which is characterized by the fact that Transact-SQL, which is a development of the ANSI/ISO standard, is used as the query language.
MongoDB	mongoDB.	It is an open-source NoSQL database that is document-oriented. MongoDB uses JSON like documents to store any data. It is written in C++.
R Server	R	RStudio Server enables to provide a browser based interface to a version of R running on a remote Linux server, bringing the power and productivity of the RStudio IDE to server-based deployments of R.

4.3 Frontend

PRODUCT		USE
HTML5		Language used to create and present web pages.
Scala		A general-purpose programming language providing support for functional programming and a strong static type system.
JavaScript		A scripting programming language, created by Netscape, most often used on websites.
TypeScript		A free and open-source programming language created by Microsoft as a superset of the JavaScript language. It enables optional static typing and object-oriented class-based programming.

4.4 Security and testing

PRODUCT		USE
Owasp ZAP		Multifunctional tool for testing the security of web applications.
Kali Linux		Linux distribution intended for penetration testing. Includes tools for monitoring network traffic, collecting information about systems, networks, system configurations, password cracking and vulnerability search.
SQL map		A script written in Python to support penetration tests of web applications - allows searching for SQL Injection vulnerabilities.
nmap		Open-source tool for network exploration and security audits. Designed to quickly scan large networks, but also works well for individual addresses. Nmap uses low-level IP packets to

		detect which addresses are available on the network, what services they provide (application name and version), what operating systems they work on (system version), what types of firewall systems are used and dozens of other features.
Jmeter		A tool to automate performance testing of web applications.
Soup UI		Software for testing applications that use SOAP protocol (Simple Object Access Protocol) consisting in the exchange of information between applications using WebServices communicating through messages in XML format. Supports functional, load and safety tests.
Log4j		Java programming language library used to create logs during application operation.
Log4net		.Net library for creating logs while the application is running.

5. Conclusions

1. Creating a system architecture requires the use of generic statistical models, including GSIM and GSBPM, which describe statistical production processes, data processing and metadata definitions.
2. In the course of analysing the system's conceptual assumptions, 10 key requirements for the IT tool were defined, that will ensure integrated access to information necessary from the point of view of tourism statistics.
3. The system prototype assumes the use of innovative micro-services such as Visual Modeler, Linkage & Imputation, Disaggregation and Flash Estimates created as part of the WPJ package that will be an integral part of the prototype.
4. The release and testing of selected micro-services were promising. All countries have tested at least one of the system module/component. The acceptance and adoption for production purposes is an undergoing task. Follow-up and control is currently being performed.
5. The prototype micro-services have been deployed for testing and acceptance in three phases.
   a) Phase 1 : Testing 1 (2019Q1)
      In this exploratory phase, the first component deployed was the scraper for the hotel.com site belonging to Expedia.
   b) Phase 2: Testing 2 (2019Q2 / Q3)
      Three new components are released in this phase, namely the Booking.com scraper (v1), the Geo-localization (v1) tool and the Visual Modeler (v1). The release of the hotel.com (v2) is also execute in this phase.
   c) Phase 3: Testing 3 and Acceptance 1 (2019 Q4 and 2020 Q1)
      Hotel.com (v3), Geo-toll (v2) and Visual Modeler (v2) were released, tested and accepted with some remarks. Also in this phase the R-analysis and integration component has been released. It contains three micro-services: Linkage & Imputation, Disaggregation and Flash Estimates.
6. The initial implementation of the concept will be presented in final deliverable.

References

Buuren, S. V., & Groothuis-Oudshoorn, K., (2011). Mice: Multivariate Imputation by Chained Equations in R. Journal of Statistical Software.

Camino R. D., Hammerschmidt C. A., State R., (2019). Improving Missing Data Imputation with Deep Generative Models, arXiv:1902.10666 [cs.LG].

Dempster, A. P., Laird N.M., Rubin D. R., (1977). Maximum Likelihood Estimation from Incomplete Data Via the EM Algorithm. Journal of the Royal Statistical Society, Series B 39, pp. 1-38.

Generic Statistical Business Process Model GSBPM (2019). Available through: https://statswiki.unece.org/display/GSBPM/GSBPM+v5.1 – [Accessed 29.05.2020]

GSIM and standards (2013). Available through: https://statswiki.unece.org/display/gsim/GSIM+and+standards – [Accessed 29.05.2020].

https://statswiki.unece.org/display/gsim/Generic+Statistical+Information+Model – [Accessed 29.05.2020].

https://statswiki.unece.org/display/GSBPM/IV.+Levels+1+and+2+of+the+GSBPM – [Accessed 29.05.2020].

ESSnet WP F (2019a). "WPF - Process and Architecture", presentation at the ESSnet Big Data Pilots II, December 2019, Vienna. Cooperation between ISTAT (Statistics Italy), NSI (Statistics Bulgaria), Destatis (Statistics Germany), DST (Statistics Denmark), Statistics Estonia, INSEE (Statistics France), CBS (Statistics Netherlands) and INE (Statistics Portugal). Available through: https://webgate.ec.europa.eu/fpfis/mwikis/essnetbigdata/images/5/5f/Implementation_Track_Meeting_2019_12_09-10_Vienna_Presentation_WPF.pdf – [Accessed 29.05.2020].

ESSnet WP F (2019b). BREAL - Big Data REference Architecture and Layers and Big Data Life Cycle for Official Statistics, Deliverable F1. Available through: https://webgate.ec.europa.eu/fpfis/mwikis/essnetbigdata/images/6/65/WPF_Deliverable_F1_BREAL_Big_Data_REference_Architecture_and_Layers_v.03012020.pdf – [Accessed 29.05.2020].

ESSnet WP J (2020). Interim technical report showing the preliminary results and a general description of the methods used, Deliverable J2. Available through: https://webgate.ec.europa.eu/fpfis/mwikis/essnetbigdata/images/6/60/WPJ_Deliverable_J2_-_Interim_technical_report_showing_the_preliminary_results_and_a_general_description_of_the_methods_used_2020_01_07.pdf – [Accessed 29.05.2020].

Annexes

The Netherlands – data analysis

Flash estimates:

The model has run for several large cities and NUTS2-regions. In the figures below you can see a strong drop from March onward, corresponding to fewer tourists due to COVID-19. For Amsterdam, it looks plausible. For Den Haag (The Hague), the values become negative.

Figure 7. Flash estimates of the number of quests in Amsterdam in 2018-2020

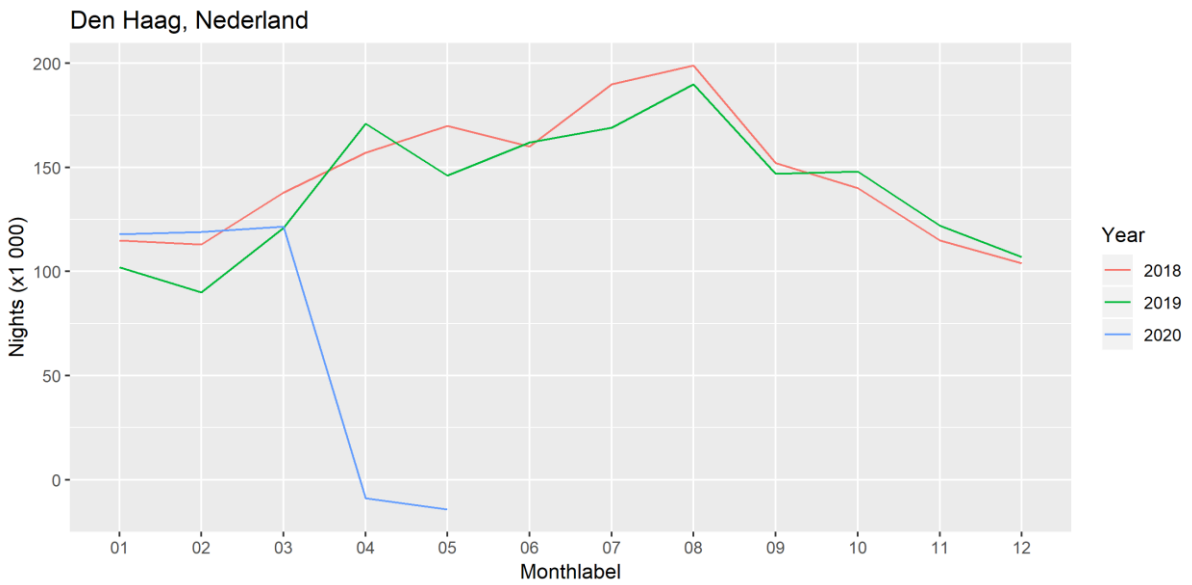

Figure 8. Flash estimates of the number of nights spent in Amsterdam in 2018-2020

In the figure below, the blue dots are the training data. The orange dots are the forecasts. For March, the mean price was around 400, close to the range of prices in the training data. For April and May, the prices were extremely low, around 100 and 200. One could argue that the model is not valid for this price range that is far away from the training data. The forecasts could improve once survey data for March and April will be available (that is, once blue dots in the 100 – 200 price range occur).

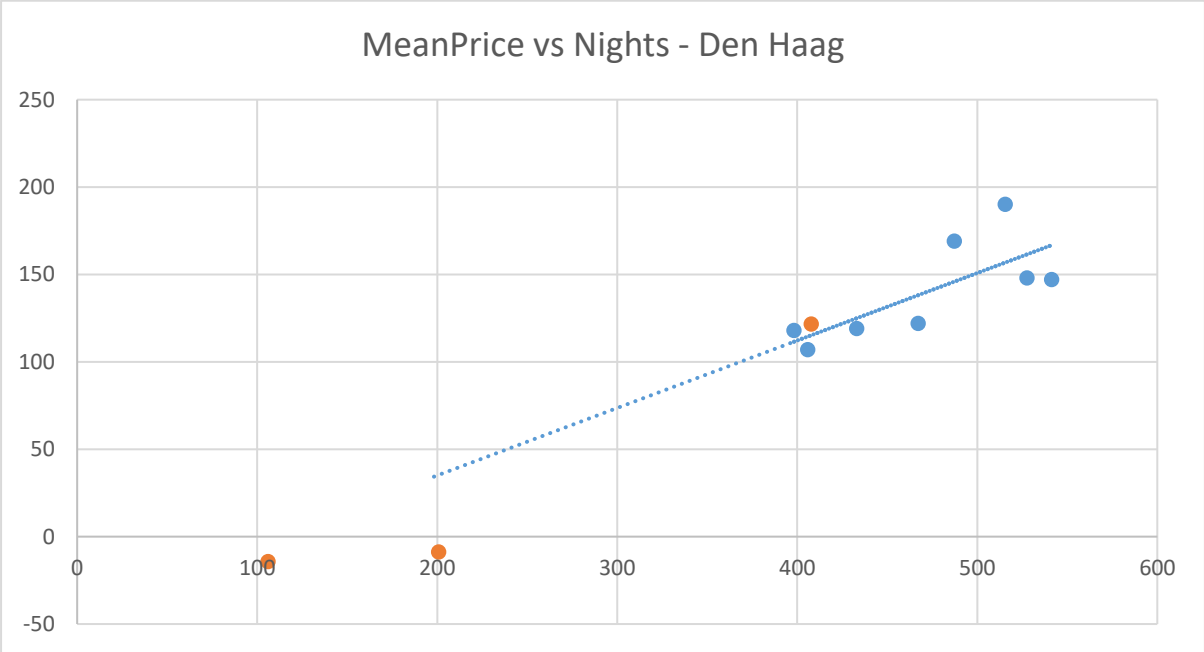

Figure 9. Difference between training data and forecasts for mean prices of accommodation establishments in Hague

Temporal disaggregation:

The provided script was first run with the data from Poland, then with data on Amsterdam. See the figure below.

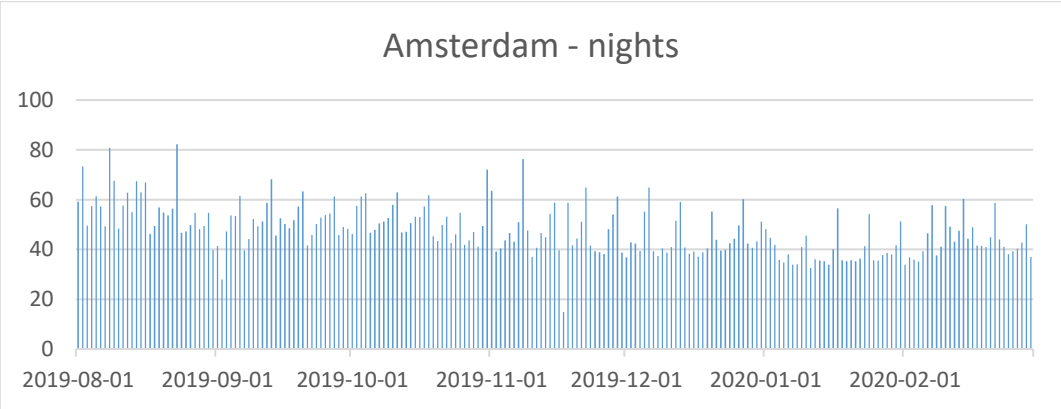

Figure 11. Temporal disaggregation of nights spent in Amsterdam

Peaks occur on Fridays. These could be people booking a weekend in Amsterdam. The disaggregation shows the busiest day for *starting* a holiday. However, it does not show how long the tourists are staying. One would expect a lot of tourists staying from Friday night to Sunday morning. The

disaggregated data will only show Friday. Thereby underestimating the number of tourists on Saturday. The disaggregated data shows a very low point on November 17th. This corresponds with a very high number of offers in the web scraping data for that day.

For some variables, the value is the same for every day, see the table below. This also occurred when the data from Poland were used, but in that case it was a different set of variables being the same. The auxiliary file on R2 shows a zero for these variables.

	A	B	C	D	E	F	G
1	Date	Tourist	"Polish	Foreigr	Nights	Polish I	Foreigr igh
2	1-8-2019	26,06452	4,225806	21,87097	59,14132	8,032258	50,32017
3	2-8-2019	26,06452	4,225806	21,87097	73,32955	8,032258	64,57338
4	3-8-2019	26,06452	4,225806	21,87097	49,55933	8,032258	41,75992
5	4-8-2019	26,06452	4,225806	21,87097	57,36317	8,032258	49,66118
6	5-8-2019	26,06452	4,225806	21,87097	61,3883	8,032258	53,6743
7	6-8-2019	26,06452	4,225806	21,87097	57,25598	8,032258	49,35625
8	7-8-2019	26,06452	4,225806	21,87097	49,29824	8,032258	41,07778

Figure 12. The output of the script of temporal disaggregation

Building user-friendly and flexible solutions in R

The System for Tourism Statistics in the field of statistical methods may be based, among others, on scripts written in R. In order to create solutions that are user-friendly and do not require advanced programming skills in R, but can be easily disseminated among partner institutions, two approaches can be used:

- Creating scripts with your own functions
- Creating packages that contain their own functions

In the first attempt, the script user has an insight into the functions that is used to generate results. An advanced user can modify the functions for own needs, but by definition the script is written for people with little knowledge of the R language. Such a function should provide parameters controlling the method performed by the script and parameters that allow manipulation of input data, which saves time preparing input files. It is best if the function is written in a way that it can be called immediately after loading the input files with default parameters and then immediately save the results.

The figure 13 below presents a fragment of a script for disaggregating monthly data on the use of accommodation base for daily data based on a series of daily data from accommodation portals.

Temporal Disaggregation.R

Source on Save

```
1   library('lubridate')
2   library("dplyr")
3   library(readr)
4   library(lars)
5
6   # parameters
7   TD_lasso = function(survey.m,
8                       portal.d,
9                       trimming = 'automatic',
10                      start_date = NULL,
11                      end_date = NULL,
12                      varX = c('offers','supply','mean_price'),
13                      benchmarking_method = 'multiplicative'
14                      )
15  {...}
105
106 # Generating outputs
107 # 1 - disaggregated variables
108 # 2 - R2
109 setwd('//vmfrze01/ESSnet BDII-WPJ/Zbiory danych')
110 survey.m <- read_delim("KT1_szeregi/Survey_Monthly statistics - Województwo mazowieckie, Polska.csv",";")
111 portal.d <- read_csv2("Hotels do R/Results/Daily statistics - Województwo mazowieckie, Polska.csv")
112
113 TD <- TD_lasso(survey.m,portal.d)
114 TD[[1]] %>% write.csv2("Dezagregacja/Temporal/Disaggregated data - Województwo mazowieckie, Polska.csv")
115 TD[[2]] %>% write.csv2("Dezagregacja/Temporal/R2 - Województwo mazowieckie, Polska.csv")
116
```

Figure 13. Sample code of disaggregation data script

In lines 1-4, packages created by other members of the R community are loaded. Then there is the definition of the time disaggregation function along with the default parameters of this function. In the lines 15-105 a body of functions is presented that the user can view, but does not need it to generate the results. The description of the function and parameters controlling it can be placed in the form of comments in the script itself or in external documents. In lines 109-111, the working folder is set, batch files are loaded, and in lines 113-115, the function is called and the results are saved (disaggregated data and model statistics). It is worth noting that after setting the working folder and batch file names, the script can be called in it's entirely without any further intervention on the part of the user.

The second approach is to create packages that contain their own functions. The package is a richer creation than the script itself, because it contains files describing the functions, examples of their use and even sample data sets. In this situation, the user does not have easy access to the function code and cannot change it from the executable script level. The following figure shows a fragment of a script based on its own package.

```
Temporal Disaggregation.R*
Source on Save
1  library("StatFunctions")
2  library('lubridate')
3  library("dplyr")
4  library(readr)
5  library(lars)
6
7  # Generating outputs
8  # 1 - disaggregated variables
9  # 2 - R2
10 setwd('//vmfrze01/ESSnet BDII-WPJ/Zbiory danych')
11 survey.m <- read_delim("KT1_szeregi/Survey_Monthly statistics - Województwo mazowieckie, Polska.csv",";")
12 portal.d <- read_csv2("Hotels do R/Results/Daily statistics - Województwo mazowieckie, Polska.csv")
13
14 TD <- TD_lasso(survey.m,portal.d)
15 TD[[1]] %>% write.csv2("Dezagregacja/Temporal/Disaggregated data - Województwo mazowieckie, Polska.csv")
16 TD[[2]] %>% write.csv2("Dezagregacja/Temporal/R2 - Województwo mazowieckie, Polska.csv")
```

Figure 14. Sample code for creating package

The first line loads its own "StatFunctions" package containing the TD_lasso function called in the script. After loading the remaining packages, the user immediately proceeds to loading the startup files, calling the function and saving the results. As part of the packages, files are created describing the function. Below is an example of the read_delim function description from the readr package.

Files | Plots | Packages | Help | Viewer

R: Read a delimited file (including csv & tsv) into a tibble | Find in Topic

read_delim {readr} R Documentation

Read a delimited file (including csv & tsv) into a tibble

Description

`read_csv()` and `read_tsv()` are special cases of the general `read_delim()`. They're useful for reading the most common types of flat file data, comma separated values and tab separated values, respectively. `read_csv2()` uses `;` for the field separator and `,` for the decimal point. This is common in some European countries.

Usage

```
read_delim(file, delim, quote = "\"", escape_backslash = FALSE,
  escape_double = TRUE, col_names = TRUE, col_types = NULL,
  locale = default_locale(), na = c("", "NA"), quoted_na = TRUE,
  comment = "", trim_ws = FALSE, skip = 0, n_max = Inf,
  guess_max = min(1000, n_max), progress = show_progress(),
  skip_empty_rows = TRUE)
```

Figure 15. View of the function description directly in the programming environment